Evolutionary Stress Tests Reveal Parent-Locked Tonic Modulation in Neural Cellular Automata

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Abstract

We set out to test whether closed-loop control of global chemical signals makes regenerative Neural Cellular Automata (NCAs) more robust, and built the evaluation to prove it: recurring multi-block lesions larger than the perception radius, a small evolved controller for three broadcast modulator channels, held-out damage seeds. On a single trained model the experiment looked like a success. Crossing five independently trained parent seeds — two independent controller evolutions each, under a preregistration fixed before data collection — dissolves that success into three findings. First, the robustness we had attributed to control is mostly a property of training: parents trained with channels present beat unmodulated siblings in five of five seeds (median final-Hamming reduction 0.008, up to ≈ 0.14 on the most fragile parent) with modulation pinned to neutral. Second, evolved controllers add little on top: on four parents the effect is absent or noise-level; on the fragile parent both evolutions reproducibly help (0.035–0.038 $\rightarrow$ 0.029–0.030, survival 0.975 $\rightarrow$ 1.00) — yet its tonic vector is nearly identical (cosine 0.99) to a sibling's that does not help, so the benefit is an interaction with parent dynamics, not a distinctive operating point. All ten controllers emit flat, nonzero, parent-specific constants with no lesion-locked response. Third, these constants are organism-locked: a single-donor probe penalizes five of five siblings (lethally in two), a full 5×5 transfer matrix shows most transfers harmful and none better than the recipient's own controller, and injecting each donor's tonic constant directly — no controller at all — reproduces its transfer outcomes in 46 of 50 cells, including every lethal one. Single-parent evaluation would have approved all of it. The attribution protocol (adversarial damage calibration, hacking and initialization probes, parent-seed-resolved decomposition) is the transferable artifact.

1 Introduction

Growing Neural Cellular Automata (GNCAs) grow a target morphology from a single seed and regenerate it after damage using one local, differentiable update rule [1]. The catch is perception: a cell sees only its immediate neighborhood, so wounds larger than a few cells contain tissue with no signal to integrate, and a severed fragment has no way to know what it was part of. The natural remedy is a global signal. Prior work, our own single-parent study included (Appendix H), adds global modulator channels and evolves release policies for them — and reports the outcome on one trained model.

That single-model habit is the problem this paper is about. When a modulated NCA outperforms an unmodulated one, three effects are indistinguishable in the report: the channels being *present during training*, the controller *acting at run time*, and whatever *transfers* to another organism. Our single-parent study exhibited the failure firsthand: it credited closed-loop control for a 2× improvement that a later pilot traced elsewhere — the evolved controller transferred lethally to sibling parents while channel parents with modulation pinned to zero beat the unmodulated baseline in every seed (Appendix C). The improvement was real; the attribution was wrong.

We therefore ran the attribution study that the original design needed: preregistered, with the primary statistic and decision thresholds fixed before data collection. Five parent seeds, each trained from scratch as a K=0 pair and a channel-aware K=3 pair; *two* independent controller evolutions per channel parent; four conditions per seed — $H(K{=}0, s)$, $H(K{=}3, m{=}0, s)$, $H(K{=}3, \text{own ctrl}, s)$, and a cross-parent probe — all on held-out damage seeds; then, post-hoc, a full 5×5 transfer matrix between all ten controllers and all five parents, and a tonic-transplant condition that replaces each controller with its realized mean output (Section 3). Preregistrations and the mechanical analysis scripts are in the repository history, timestamped before the runs they govern.

Contributions.

1. A preregistered robustness-attribution protocol for regenerative NCAs: adversarial damage calibrated to exceed perception, crossed with parent-seed variation, decomposed into E_{train}, E_{ctrl}, E_{transfer}, with calibration probes that keep the evolution landscape interpretable (Appendix F).
2. The attribution, settled causally. Training with channels is the robust cause (5/5 seeds). The controller effect is parent- dependent — absent or noise-level on four parents, reproducibly present on the fragile one — and the evolved artifact is a parent-specific tonic constant, not a policy: it has no lesion-locked response, it penalizes every sibling it meets (lethally in two of five), and injecting the constant alone reproduces the controller's transfer outcomes, every lethal one included. The transfer matrix (23/40 off-diagonal cells harmful, none better than the recipient's own controller) closes the case.

2 Related Work

Neural cellular automata. GNCAs grow patterns from a seed and regenerate after damage [1], with extensions to self-classification [2] and textures [3]. All rely on local perception, and regeneration degrades beyond the perception radius — the regime our benchmark exploits. Signal channels [4], goal conditioning [5], and information-dynamical analyses of self-maintenance [6] all treat the signal as fixed input or emergent byproduct, never a controlled output. We ask the prior question: when such a channel *appears* to help, what is actually helping?

Search over evaluation regimes. Our design follows the insight, central to novelty search [7] and quality-diversity optimization [8], that a fixed objective can misdirect search and hide failure modes. Environment-generation methods co-evolve challenges with solutions: POET [9], PAIRED [10], ACCEL [11]. Our benchmark does not yet evolve the damage distribution; it is the controlled stationary baseline against which such co-evolution can be judged.

3 Experimental design

Damage regime and parents. Recurring multi-block lesions — every 150 steps, n=4 contiguous 16×16 blocks at seeded positions, T=2000 — exceed the perception radius, so wound interiors have no living neighbors (the information-isolation regime mapped in Appendix E). Damage seeds 0–7 drive evolution; held-out seeds 10000–10007 drive all reported numbers; the schedule is deliberately adversarial (Appendix F). Per seed s we train a K=0 and a channel-aware K=3 parent from scratch (Appendix D); the K=3 parent's three global modulator channels are the only non-local pathway. A 259-parameter controller (4⟶32⟶3, tanh) reads four target-free grid statistics and sets release every 10 steps; each parent gets *two* independently evolved controllers via CMA-ES [12] in Evosax [13] (population 64, σ_0=0.01, 300 generations, different evolution seeds, event-weighted Hamming objective; Appendix G).

Preregistered comparison. All conditions run on the same held-out damage seeds (5 condition seeds × 8 damage seeds) from a shared t=0 state grown by the K=3 parent. The primary statistic is

$$\Delta_s \;=\; H(K{=}3, m{=}0, s) \;-\; H(K{=}3, \text{own controller}, s), \tag{1}$$

positive when the evolved controller helps *its own* parent. Fixed before data collection: *controller effect supported* if $\Delta_s > 0$ in $\geq 4/5$ seeds with sign-consistent differences; *channel-training effect supported* if $H(K0, s) - H(m0, s) > 0$ in $\geq 4/5$; *transfer failure supported* if the July controller underperforms the own-controller substantially in $\geq 4/5$. Effects are reported per seed with median and range; we make no significance claims at five seeds. The defense study reported here fixed the two-evolutions-per-parent rule in its preregistration before launch (`experiment_results/20260818_evoseed_defense/PREREGISTRATION.md` in the repository history). Controller-output (m_t) series distinguish tonic calibration from event-locked policy (Appendix B). A post-hoc follow-up extends the single-donor July probe to the full 5×5 transfer matrix among the five defense parents, using both controller replicas per parent (evaluation only, 8 held-out damage seeds × 3 condition seeds per cell; Table 4), plus a *tonic transplant* condition that injects each donor's realized mean m_t as a constant in place of its controller.

4 Results

Table 1 reports the preregistered decomposition; the full per-seed, per-condition numbers behind it are in Appendix A.

Table 1: Final Hamming (mean over 5 condition seeds × 8 held-out damage seeds) per parent seed: the K=0 parent; the channel-aware K=3 parent with modulation pinned to zero; and that parent's own evolved controller (two independent evolutions; range where the runs differ). Survival shown in parentheses where it departs from 1.00. Channel-aware training helps in 5/5 seeds; the controller effect is parent-dependent.

Parent seed	K=0	K=3, m=0	Own ctrl (2 runs)	Interpretation
0	0.035–0.036	0.029	0.029–0.030	channel effect; no control effect
1	0.163–0.175 (surv. 0.05–0.08)	0.035–0.038 (surv. 0.975)	0.029–0.030 (surv. 1.00)	channel effect + replicated tonic benefit
2	0.030	0.018–0.021	0.022–0.026	channel effect; controller slightly worse
3	0.049–0.052	0.028–0.029	0.032–0.033	channel effect; controller slightly worse
4	0.042–0.045	0.033	0.034–0.035	channel effect; no control effect

Outcome. Applying the preregistered rule: the *channel-training effect is supported* ($H(K0, s) - H(m0, s) > 0$ in 5/5 seeds; bar was ≥4/5); the *transfer failure is supported* (penalty in 5/5; lethal in 2). The controller effect is classified as *parent-dependent* — preregistered **Outcome C**: absent or noise-level on most parents, but reproducibly beneficial on the fragile parent — on seeds 0, 2, 3, and 4 the own controller matches zero-output or is slightly worse, while on seed 1 both independent evolutions reduce final Hamming from 0.035–0.038 to 0.029–0.030 and raise survival from 0.975 to 1.00.

What the controller actually emits. All ten evolved controllers emit a *constant at a nonzero level*: within-rollout per-channel std of m_t is 0.0002–0.0046 — tonic, with no lesion-locked response, in 10/10 runs — while channel means sit at parent-distinct offsets (Appendix B, Figure 2). Evolution neither collapses to neutral output nor discovers a policy: it finds a parent-specific tonic calibration. The tonic vector alone does not predict benefit: seeds 1 and 4 emit nearly identical tonic vectors (cosine 0.993), yet only seed 1 benefits. The benefit arises from the interaction between a tonic calibration and parent-specific local dynamics, not from a distinctive chemical operating point.

Cross-parent transfer matrix. Extending the single-donor probe to the full 5×5 matrix among the five defense parents — both controller replicas per parent, three condition seeds per cell, 40 off-diagonal cells (Table 4, Figure 1) — 23/40 transfers are harmful (survival < 0.9 or final Hamming worse than the recipient's zero-output baseline by > 0.005), 15 are indistinguishable from zero-output, and 2 beat it. No foreign controller beats the recipient's own controller beyond noise. Lethal transfer replicates across controller replicas and condition seeds: donor s3 kills recipient s2 in both replicas (survival 0.00); donor s0 on recipient s1 is lethal in e1 (0.00) and near-lethal in e2

(0.25). Classifications are stable across condition seeds — only marginal cells within ±0.005 of the threshold flip. All lethal instances pair strongly negative tonic-vector cosines (s3→s2: −0.61; s0→s1: −0.93), while the only beneficial transfers — beating zero-output and matching the own controller — occur exclusively within the tonic-aligned pair s4→s1 (cosine +0.99, both replicas). Across all 40 cells, donor–recipient tonic cosine correlates with transfer penalty (Pearson $r = -0.30$) and survival ($r = +0.34$); lethal and beneficial transfers occupy opposite alignment regimes, though alignment does not by itself determine outcome (some benign cells have negative cosine, down to −0.95).

Tonic transplant. Injecting each donor's realized mean m_t as a constant in place of its controller (both replicas, three condition seeds) reproduces the full controller's outcome in 36/40 off-diagonal cells, including every lethal transfer (the four mismatches are magnitude differences inside already-lethal cells), and matches the recipient's own controller in all 10 diagonal cases (46/50 overall; Appendix A). The transferable component is thus the tonic setpoint; the controller's small dynamic residual did not measurably contribute, on-parent or off.

5 Discussion

Which component causes robustness. Channel-aware training is the broad robustness effect: zero-output channel parents beat their K=0 siblings in 5/5 seeds, with the largest effect exactly where the unmodulated parent is most fragile (seed 1: 0.163–0.175 → 0.035–0.038; survival 0.05–0.08 → 0.975). Evolved modulation is parent-dependent: on four parents the own controller matches zero-output or is slightly worse; only on the fragile parent do both independent evolutions add a final stabilization (survival 0.975 → 1.00, final Hamming → 0.029–0.030). Robustness thus resides in the interaction of parent training and global state, not in a transferable closed-loop policy.

Why cross-parent transfer fails — and when it does not. The full matrix (Table 4) sharpens the single-donor probe: transfer is not universally lethal, but it is parent-locked in a precise sense. No foreign controller beats the recipient's own controller; the majority of transfers are harmful (23/40); lethal failures replicate across controller replicas and condition seeds; and the only beneficial transfers occur within the tonic-aligned pair (s4→s1) (Section 4). The tonic transplant makes the mechanism causal: injecting the donor's mean m_t as a constant reproduces the controller's outcome in 36/40 off-diagonal cells (10/10 diagonal), including every lethal one — what transfers, for good or ill, *is* the constant. Each controller's output is a calibration against its own parent's channel weights, so the same release level drives different dynamics in a sibling. Evolution found an operating point for one organism, not a modulation law.

Limitations and future work. Five parent seeds; two controller evolutions per parent; one morphology; one stationary damage family; the July transfer probe remains single-donor. Only one of five parents proved controller-responsive, so the prevalence of such parents is unknown. Next: multi-parent (population-based) evolution; full transfer matrices; non-stationary, diversity-driven damage co-evolution, for which this benchmark is the controlled baseline.

6 Conclusion

In regenerative NCAs under adversarial recurring damage, robustness comes from channel-aware training (5/5 seeds, rescuing a nearly lethal unmodulated parent); evolved modulation is a parent-dependent tonic calibration — noise-level on most parents, beneficial on the fragile one, parent-locked under transfer. The broader contribution is methodological: attribute robustness claims across model seeds, not only damage seeds.

A Full per-seed, per-condition results

Table 2: Final Hamming (mean ± SD, 5 condition seeds × 8 held-out damage seeds) and survival for each of the ten runs (5 parent seeds × 2 independent controller evolutions) under hard recurring multi-block damage, T=2000. Conditions: K=3 parent with modulation pinned to zero; the parent's own evolved controller; the K=0 parent.

	K=3, m=0		K=3, own ctrl		K=0	
Run	H	surv	H	surv	H	surv
s0_e1	0.0290±.0135	1.000	0.0303±.0134	1.000	0.0361±.0137	1.000
s0_e2	0.0294±.0144	1.000	0.0292±.0136	1.000	0.0354±.0109	1.000
s1_e1	0.0384±.0266	0.975	0.0286±.0177	1.000	0.1628±.0389	0.075
s1_e2	0.0353±.0256	0.975	0.0301±.0210	1.000	0.1746±.0356	0.050
s2_e1	0.0183±.0145	1.000	0.0225±.0149	1.000	0.0299±.0178	1.000
s2_e2	0.0208±.0170	1.000	0.0258±.0161	1.000	0.0296±.0174	1.000
s3_e1	0.0281±.0177	1.000	0.0332±.0165	1.000	0.0491±.0084	1.000
s3_e2	0.0289±.0182	1.000	0.0321±.0164	1.000	0.0525±.0136	1.000
s4_e1	0.0332±.0233	1.000	0.0336±.0223	1.000	0.0420±.0111	1.000
s4_e2	0.0332±.0225	1.000	0.0345±.0222	1.000	0.0452±.0093	1.000

Table 3: Cross-parent transfer probe (unchanged from the per-parent study): the July donor controller evaluated on each recipient parent. Final Hamming and survival on held-out damage seeds; transfer is a penalty in 5/5 recipients, lethal in 2.

Recipient seed	Final Hamming	Survival
0	0.036	1.00
1	0.249	0.00 (lethal)
2	0.434	0.00 (lethal)
3	0.100	0.45
4	0.041	1.00

Tonic transplant. Replacing each donor controller by its realized mean m_t injected as a constant (same protocol, both replicas, 3 condition seeds) reproduces the full controller's transfer outcome in 46/50 cells (final Hamming within 0.01 and same survival class), including every lethal cell — e.g. s3→s2: controller 0.208/0.00 vs transplant 0.186/0.12 (e1) and 0.239/0.00 vs 0.234/0.00 (e2); s0→s1: 0.470/0.00 vs 0.353/0.00 (e1). The four mismatches are magnitude differences inside already-lethal cells. A recipient's own tonic transplant is likewise indistinguishable from its own controller in all ten diagonal cases. In total, the transplant reproduces the controller's outcome in 36/40 off-diagonal cells and 10/10 diagonal cells (46/50 overall). Under the tested regime, the small dynamic residual of the controllers did not measurably contribute, on-parent or off (full CSVs in the repository, `experiment_results/20260822_transfer_matrix/cs3_*`).

AUC mirrors the same ordering in every seed (full CSVs in the repository).

B Controller-output (m_t) diagnostics

The tonic vector alone does not predict benefit: seeds 1 and 4 emit nearly identical tonic vectors (cosine similarity 0.993, Euclidean distance 0.0065), yet only seed 1 benefits from its controller. The benefit arises from the interaction between a tonic calibration and parent-specific local dynamics, not from a distinctive chemical operating point.

Table 4: Full 5×5 cross-parent transfer matrix among the five defense parents, both controller replicas (top: e1; bottom: e2). Rows: donor controller; columns: recipient parent. Cells give mean final Hamming / worst-case survival over 3 condition seeds × 8 held-out damage seeds (24 rollouts per cell). Diagonal (bold): recipient's own controller (survival 1.00 throughout). †lethal (survival 0.00). Across the 40 off-diagonal cells: 23 harmful, 15 indistinguishable from zero-output, 2 beneficial — the beneficial cells are exclusively s4⟶s1, the tonic-aligned pair (cosine +0.99), in both replicas.

e1 donor	recip. s0	recip. s1	recip. s2	recip. s3	recip. s4
s0	**0.0315**	0.470/0.00†	0.097/0.38	0.028/1.00	0.052/0.88
s1	0.035/1.00	**0.0306**	0.023/1.00	0.049/1.00	0.035/1.00
s2	0.029/1.00	0.043/0.88	**0.0201**	0.033/1.00	0.092/0.50
s3	0.035/1.00	0.036/0.88	0.208/0.00†	**0.0295**	0.044/1.00
s4	0.034/1.00	0.030/1.00	0.023/1.00	0.044/1.00	**0.0340**
e2 donor					
s0	**0.0291**	0.108/0.25	0.056/0.88	0.032/1.00	0.039/1.00
s1	0.032/1.00	**0.0292**	0.025/1.00	0.049/1.00	0.032/1.00
s2	0.041/1.00	0.074/0.62	**0.0219**	0.033/1.00	0.113/0.25
s3	0.028/1.00	0.058/0.75	0.239/0.00†	**0.0262**	0.028/1.00
s4	0.030/1.00	0.030/1.00	0.020/1.00	0.033/1.00	**0.0336**

Table 5: Tonic output of the evolved controllers: per-channel mean of m_t, averaged over the two independent evolutions per seed. All ten controllers sit at nonzero, parent-distinct offsets; within-rollout per-channel std of m_t is 0.0002–0.0046 everywhere — flat, tonic output with no lesion-locked response in 10/10 runs.

Seed	mean(m) per channel (avg. of 2 runs)
0	−0.021 +0.002 +0.027
1	+0.030 +0.012 −0.026
2	−0.026 −0.032 −0.015
3	+0.001 +0.032 −0.010
4	+0.028 +0.007 −0.025

C Pilot: single-parent evaluation hides parent-locking

A three-parent pilot (2026-08-16) first exposed the confound. The single-parent controller from our original five-condition study transferred lethally to two of three sibling parents (survival 0.00, final Hamming 0.249 and 0.434), while zero-output channel parents beat the $K=0$ baseline in 3/3 seeds (0.029/0.034/0.020 vs 0.034/0.177/0.028). The original table's headline gap was therefore part parent-training effect and part parent-seed luck — motivating this study.

D Model and training details

The grid state $x_t \in [0,1]^{96\times 96\times 16}$ holds 16 channels; the last four are RGBA with alpha last. Perception uses three fixed kernels (identity, Sobel-x, Sobel-y) per channel, giving $p_t \in \mathbb{R}^{48}$:

$$p_t = \left(K_{\mathrm{id}} * x_t,\ K_{\mathrm{S}_x} * x_t,\ K_{\mathrm{S}_y} * x_t\right). \tag{2}$$

The update MLP maps $(48{+}K)$ inputs to 128 hidden units (ReLU) and back to 16 channel increments, with the final layer zero-initialized. Cells update with probability 0.5; a cell is alive when its alpha exceeds 0.1 within its 3×3 max-pooled neighborhood; dead cells are masked to zero.

Tonic and phasic modulator dynamics:

$$m_t^{(\mathrm{tonic})} = \alpha\, m_{t-1}^{(\mathrm{tonic})} + (1-\alpha)\, c_t, \qquad m_t^{(\mathrm{phasic})} = m_{t-1}^{(\mathrm{phasic})} \cdot e^{-\Delta t/\tau}, \tag{3}$$

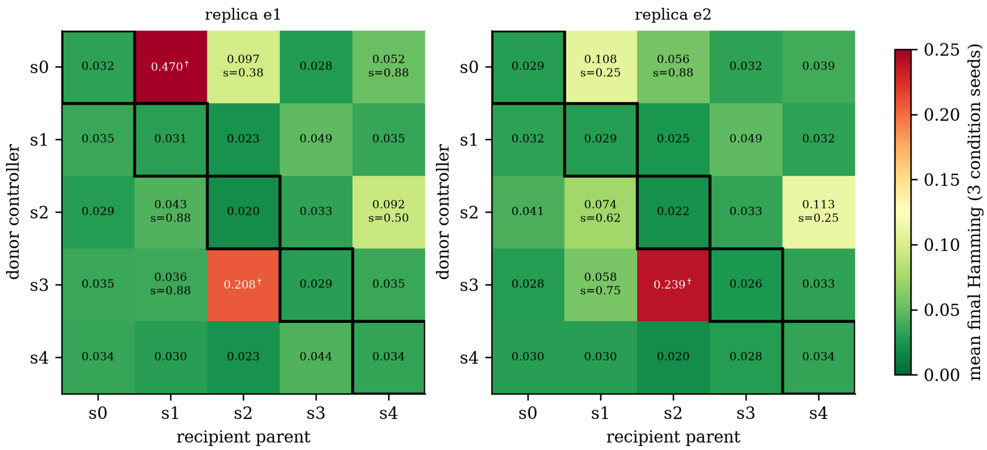

Figure 1: Cross-parent transfer matrix as a heatmap (same data as Table 4). Color: mean final Hamming over 3 condition seeds (green: matches recipient's own controller; red: catastrophic). †lethal (survival 0.00); sub-threshold survival annotated where < 1.00. Diagonal boxed: recipient's own controller. The harmful off-diagonal band and its replication across both evolutions are immediately visible; the only beneficial off-diagonal cells are s4→s1 in both replicas.

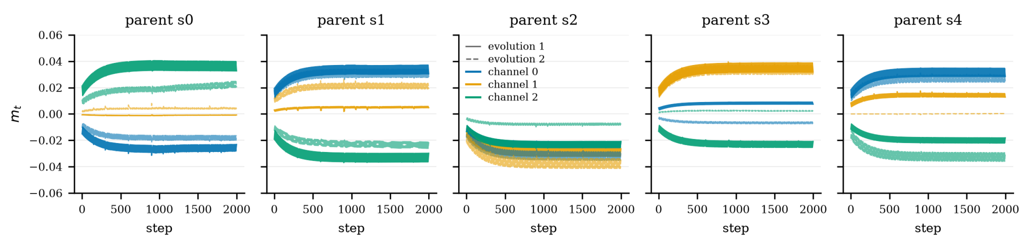

Figure 2: Controller output m_t over a full evaluation rollout, per parent seed, both independent evolutions overlaid (solid/dashed), one line per modulator channel. After a brief initial transient while the perceptual state settles, every controller holds a constant, parent-distinct offset through all lesion events — tonic calibration, not an event-locked policy. Note that the two independent evolutions of the same parent sometimes settle at slightly different setpoints.

with $\alpha = 0.95$, $\tau = 20$, and the injected level the clipped sum, broadcast to all cells and concatenated to perception (input dim $48 + 3 = 51$).

Parents train on a 40×40 emoji target zero-padded to the canvas, with pool-based training (pool 1024, batch 8, worst-in-batch reseeding), random square-erasure damage mid-episode, MSE loss on RGBA, and Adam at 10^{-3} for 8000 steps. A first attempt at 2×10^{-3} diverged to NaN near step 5532 via a late-stage loss-spike cascade; 10^{-3} absorbs the identical spike.

E Perception-radius sweep

A single-lesion sweep (radii $\{2, 4, 8, 16\}$, single- and multi-site, with and without modulator channels) maps the boundary of local repair. Recovery is near-perfect for small lesions and degrades once lesions exceed the effective perception radius. Two failure signatures motivate this work: severed fragments that survive but cannot reattach, die, or decide what to grow into, and persistent half-alpha scar tissue at wound sites. Both are information failures: the misbehaving cells are exactly the cells cut off from global context.

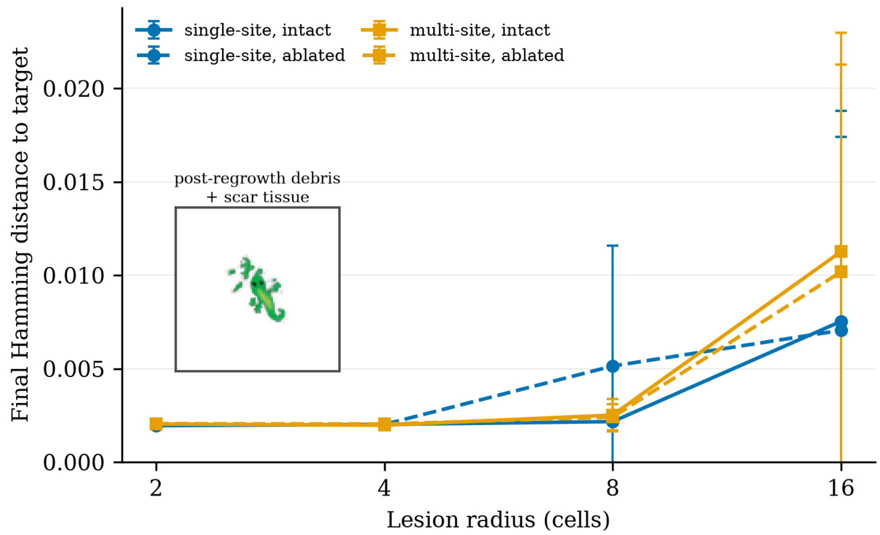

Figure 3: E1 lesion sweep. Final Hamming distance as a function of lesion radius and kind, with and without modulator channels (mean ± SD over 5 damage seeds). Inset: post-regrowth debris and scar tissue.

F Calibration probes

Collapse probe. The target mask is only 4.6% alive, so a fully-dead grid scores Hamming ≈ 0.046 — close to a struggling organism's ≈ 0.02. We verified empirically that the all-dead state scores 0.0461 against struggling-neutral's 0.0205: death loses by 2.2×, so the alive-fraction floor is disabled (0.0). Notably, the runbook-default floor of 0.3 would have *penalized every healthy candidate* (a correct lizard lives at ≈ 0.057 alive), pushing evolution toward overgrowth — a case where calibrating from the experiment's own data overrode a plausible default.

Initial step-size probe. Perturbing the neutral controller by Gaussian noise at increasing σ (25 samples each, hard-regime evaluation) showed: $\sigma \leq 0.01$ stays healthy (some samples beat neutral); $\sigma = 0.05$ is bimodal; $\sigma = 0.3$ lands *every* sample in saturated overgrowth (alive $\to$ 0.96). A first evolution run with the library-default $\sigma_0 = 0.3$ froze at fitness 0.150 from generation 2 onward — 7× worse than doing nothing — because CMA-ES was ranking overgrowth-degree noise. $\sigma_0 = 0.01$ brackets the neutral point and the same budget converged smoothly (Figure 4).

G Evolution objective

We evolve the controller with CMA-ES [12] (Evosax implementation [13]), population 64, initial step size σ_0=0.01, 300 generations, neutral (all-zero) controller as the initial mean. Fitness is the event-weighted mean Hamming distance to the target alpha mask over full rollouts on the eight train damage seeds:

$$f(\theta) = \frac{1}{\sum_t w_t} \sum_{t=1}^{T} w_t \, H_t(\theta), \qquad w_t = \exp\left(- \frac{t - \tau_{\text{last}}(t)}{\tau_w} \right), \tag{4}$$

where $\tau_{\text{last}}(t)$ is the most recent lesion step and $\tau_w = 150/3 = 50$. The recency kernel resets at each lesion, so slow repair is expensive even when endpoint Hamming is identical — it de-saturates the metric and prices repair speed directly. Fitness is not the reported metric: all results report unweighted trajectory statistics on held-out seeds. Two calibration controls make this landscape interpretable — a collapse probe (verifying the objective cannot be gamed by alpha suppression) and an initial step-size

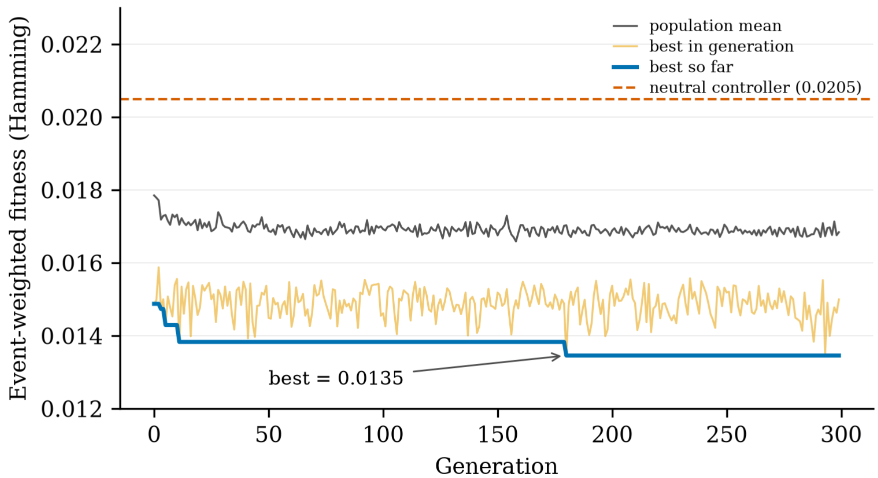

Figure 4: CMA-ES fitness over 300 generations (σ_0=0.01). Dashed line: neutral controller (0.0205). Best evolved fitness: 0.0135 — a 34% improvement with no stall.

ablation (showing σ_0=0.3 saturates the first population in overgrowth where no repair signal exists) — both detailed in Appendix F.

H Single-parent five-condition study

Five conditions share the same protocol, horizon, and damage seeds: **closed-loop** (evolved controller); **static** (the evolved controller evaluated once at t=0 and held constant — isolating temporal modulation from the evolved tonic level itself); **constant** (fixed tonic, grid-searched over $\{-1, -0.5, 0, 0.5, 1\}$ on train seeds); **random** (uniform $[-1, 1]$ each step, actuation-matched noise); and **no modulation** (K=0 parent). All experiments ran on a single rented A100; parent training took ≈20 min/model, evolution ≈2.5 h, total project cost under $15.

Modulation helps substantially (RQ1). Relative to no modulation, modulated conditions recover 2.2× more completely (final Hamming 0.028–0.030 vs. 0.063) and sustain ≈ 4× less cumulative damage (AUC 0.016–0.017 vs. 0.064). The wounds exceed the perception radius, so this improvement is attributable to the broadcast channel — the only non-local pathway — repairing damage that local rules cannot diagnose.

A regime boundary, not a failed ablation (RQ2). The three modulated conditions are statistically indistinguishable: final Hamming spans 0.028–0.030, half-life 6.4–7.2 steps, all differences within noise (pairwise effect sizes < 0.15). We read this as a measurement of the *regime*, not a failure of the controller: the damage process is stationary and memoryless — events are i.i.d. in size, number, and position — so the optimal release policy is time-invariant and a fixed tonic gain is structurally adequate. There is no temporal structure for adaptive scheduling to exploit. Evolution's measured contribution is automatic discovery of the near-optimal tonic level in 2.5 hours from a neutral start, without the hand search the constant condition required.

Random modulation is lethal. Random actuation kills every run (survival 0.00, final Hamming 0.825): pilot probes show constant levels alone swing mean Hamming from 0.003 to 0.93. The modulator channels are a high-gain control pathway, not a free architectural bonus — the identity of the release schedule determines whether the organism lives.

Metric artifacts of long-horizon evaluation (RQ3). Two artifacts surfaced only under repeated damage. First, repair half-life is defined per lesion relative to each post-lesion jump; on the baseline's chronically damaged morphology, new lesions often land on already-degraded regions, producing no measurable jump and scoring as zero half-life — so the baseline's 2.7 steps does *not* indicate fast repair. Second, residual Hamming in all conditions is partly attributable to locomotion drift during regeneration rather than permanent damage; the metric conflates spatial translation with morphological error. Both are documented so that future benchmark users can avoid them.

I Reproducibility

Parents train in ≈20–30 min each on one A100; the defense study ran two independent 300-generation controller evolutions per parent (10 total), each ≈2.5–3 h on the study A100, for a total defense-study cost of ≈$25 of rented GPU time. The transfer matrix is evaluation-only over existing artifacts: 40 off-diagonal evaluations, ≈2 minutes of A100 time per full 5×5 replica matrix. Preregistration, analysis script, per-seed artifacts (controllers, parents, trajectories, m_t series), and the mechanical outcome output are in the repository, timestamped before the results. Code will be released upon publication.

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